

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 645 Geometric Modeling — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Keenan Crane — Discrete Differential Geometry (free notes, video, and code)	Primary source. Complete lecture videos, a full set of course notes, and C++/JavaScript skeleton code. The best free resource on this subject by a wide margin.
libigl tutorial (free, with code)	A working geometry processing library with a tutorial that doubles as a survey of the field. Read the source for every algorithm you implement.
Polygon Mesh Processing — course slides (free)	Slides and the companion open-source library. Strong on simplification, remeshing, and parameterisation.
GAMES101 Lectures 11–12 — Geometry	The condensed version you met in CSCE 641. Useful as revision before Module 03.
MIT 6.8410 Shape Analysis — Justin Solomon	A graduate course covering the same operators from a more analytical direction. Good second voice for Modules 07–09.
Inigo Quilez — articles on distance functions (free)	The practical reference for implicit surfaces and SDFs, with derivations and working shader code. Primary source for Module 11.

Module-by-module map

Module 01 — Representing Shape

Resource	What it covers for this module
Keenan Crane — DDG, Lecture 1 (Overview) and the Combinatorial Surfaces lecture	Representations, manifoldness, and Euler characteristic, with the topology done carefully.
libigl tutorial — Chapter 1 (Mesh representation)	How a real library stores meshes, and why it uses flat index matrices rather than pointer structures.
Inigo Quilez — 'Distance functions'	The implicit side, with working code. Read it now for contrast; it becomes the primary source in Module 11.
Polygon Mesh Processing slides — Chapter 1	Representations compared with the operation-cost framing used here.

Module 02 — Meshes and Connectivity

Resource	What it covers for this module
Keenan Crane — DDG, Combinatorial Surfaces and the half-edge lecture	The structure, the invariants, and the traversals, with the topology made precise.
libigl tutorial — adjacency and traversal	How a production library does this with flat index arrays rather than a pointer structure, and why.
OpenMesh documentation	A mature half-edge implementation. Read the handle and circulator design before writing your own API.
Max Wardetzky et al. — 'Discrete Laplace operators: no free lunch' (free)	Not needed until Module 08, but the normal-weighting discussion here is the same kind of argument — worth seeing early.

Module 03 — Curves: Bézier and Continuity

Resource	What it covers for this module
Keenan Crane — DDG, the curves material	Bézier curves with the properties derived rather than asserted.
Freya Holmér — The Beauty of Bézier Curves (free video)	The clearest visual explanation of de Casteljau and the control properties available anywhere. Watch before reading.
A Primer on Bézier Curves (free, interactive)	Comprehensive, interactive, with derivations and working code for everything in this module including subdivision and offsetting.
GAMES101 Lecture 11 — Geometry 1 (Curves)	The condensed graphics-oriented treatment you met in CSCE 641.

Module 04 — B-Splines and NURBS

Resource	What it covers for this module
A Primer on Bézier Curves — B-spline sections (free, interactive)	Knot vectors explained interactively, which is the only way they ever make sense the first time.
The NURBS Book — Piegl & Tiller (reference)	The standard reference. Check your library; the algorithms chapter is the one to consult when implementing de Boor and knot insertion.
Freya Holmér — The Continuity of Splines (free video)	An exceptionally clear visual treatment of continuity, knot multiplicity, and the spline families compared.
OpenNURBS / Rhino developer documentation	How NURBS are represented in a real CAD kernel, including trimming.

Module 05 — Surfaces: Tensor Products and Patches

Resource	What it covers for this module
The NURBS Book — surfaces chapters	Tensor-product surfaces, trimming, and the continuity conditions, done rigorously.
Keenan Crane — DDG, surface representation material	The topological argument for why patches struggle with arbitrary surfaces.
OpenNURBS developer documentation — trimmed surfaces	How a production kernel actually represents trims and why export is difficult.
Cem Yuksel — Interactive Computer Graphics, tessellation lectures	GPU tessellation of parametric patches, connecting this module to CSCE 641.

Module 06 — Subdivision Surfaces

Resource	What it covers for this module
Keenan Crane — DDG, subdivision material	The rules derived rather than stated, with the B-spline connection made explicit.
DeRose, Kass & Truong — Subdivision Surfaces in Character Animation (free)	The Pixar paper that introduced semi-sharp creases and put subdivision into production. Short and very readable.
Jos Stam — Exact Evaluation of Catmull-Clark Subdivision Surfaces (free)	How to evaluate the limit surface directly. The eigenstructure argument is worth working through once.
OpenSubdiv (Pixar, open source)	The production implementation, with excellent documentation of feature-adaptive subdivision. Read the overview even if you do not use the library.

Module 07 — Discrete Curvature

Resource	What it covers for this module
Keenan Crane — DDG, curvature lectures and notes	The primary source for this module. The 'discretise the theorem' philosophy is developed properly, with the angle defect derivation.
Meyer, Desbrun, Schröder & Barr — Discrete Differential-Geometry Operators (free)	The paper that established the cotangent formula and the mixed area region in graphics. Short and still the standard reference.
Wardetzky et al. — Discrete Laplace operators: no free lunch (free)	The proof that no discrete Laplacian preserves all the smooth properties. Three pages, and it reframes the whole subject.
libigl tutorial — curvature	Working implementations to check your own against.

Module 08 — The Laplacian and Smoothing

Resource	What it covers for this module
Keenan Crane — DDG, the Laplace operator and smoothing lectures	The operator derived from exterior calculus, with smoothing as curvature flow.
Desbrun, Meyer, Schröder & Barr — Implicit Fairing of Irregular Meshes (free)	The paper that introduced implicit smoothing and the cotangent weights to graphics. Still the clearest account of the stability argument.
Taubin — A signal processing approach to fair surface design (free)	The $\lambda \mu$ filter, derived properly as a signal-processing problem.
libigl tutorial — Laplacian and smoothing	Working code with Eigen, including the sparse solve pattern.

Module 09 — Parameterization

Resource	What it covers for this module
Keenan Crane — DDG, conformal geometry lectures	Conformal maps, the uniformisation theorem, and why angles are the right thing to preserve.
Lévy et al. — Least Squares Conformal Maps (free)	The LSCM paper. Short, clear, and directly implementable.
Sawhney & Crane — Boundary First Flattening (free)	A fast conformal method with direct boundary control, plus a free tool to compare your results against.
Hormann, Lévy & Sheffer — Mesh Parameterization SIGGRAPH course (free)	The comprehensive survey. The reference when you need a method this module did not cover.

Module 10 — Simplification and Remeshing

Resource	What it covers for this module
Garland & Heckbert — Surface Simplification Using Quadric Error Metrics (free)	The original paper. Short, clear, and still the basis of every production simplifier.
Botsch & Kobbelt — A Remeshing Approach to Multiresolution Modeling (free)	The isotropic remeshing algorithm of §3, including the threshold choice.
Polygon Mesh Processing — simplification and remeshing chapters	The comprehensive survey with implementation detail.
Sharp, Soliman & Crane — Navigating Intrinsic Triangulations (free)	Intrinsic Delaunay: fixing operator quality without moving a single vertex.

Module 11 — Implicit Surfaces and Distance Fields

Resource	What it covers for this module
Inigo Quilez — distance functions and articles (free)	The primary practical source: exact SDFs for dozens of primitives, blending operators, soft shadows, and ambient occlusion, all with derivations and working code.
Shadertoy	Thousands of sphere-traced scenes with readable source. The fastest way to learn the idioms.
Lorensen & Cline — Marching Cubes (free summaries widely available)	The original algorithm. The lookup-table construction is worth understanding once.
Ju et al. — Dual Contouring of Hermite Data (free)	How to recover sharp features that marching cubes loses.

Module 12 — Solid Modeling and Booleans

Resource	What it covers for this module
Shewchuk — Adaptive Precision Floating-Point Arithmetic and Fast Robust Geometric Predicates (free, with code)	The paper and the freely usable implementation. The single most important reference for this module.
Jacobson et al. — Robust Inside-Outside Segmentation using Generalized Winding Numbers (free)	How to get meaningful inside/outside answers from broken meshes.
CGAL — Polygon Mesh Processing and Boolean operations	A production-quality implementation built on exact predicates. Read the preconditions section carefully.
libigl — mesh boolean tutorial	A usable boolean implementation with a short interface, built on CGAL/exact arithmetic.

Module 13 — Deformation and Shape Editing

Resource	What it covers for this module
Sorkine & Alexa — As-Rigid-As-Possible Surface Modeling (free)	The ARAP paper. Short, clearly written, and directly implementable from the text.
Sorkine et al. — Laplacian Surface Editing (free)	The linear method and the rotation problem, by the people who developed it.
Keenan Crane — DDG, the deformation material	Deformation framed in terms of the operators developed through the course.
libigl tutorial — deformation chapter	Working ARAP, biharmonic, and cage-based deformation to compare against your own.